Process Costing

Learning Objectives

1. Discuss the uses of a process cost system and how it compares to a job order system.
2. Explain the flow of costs in a process cost system and the journal entries to assign manufacturing costs.
3. Compute equivalent units.
4. Complete the four steps to prepare a production cost report.
5. Explain just-in-time (JIT) processing and activity-based costing (ABC).

LEARNING OBJECTIVE 1

Discuss the uses of a process cost system and how it compares to a job order system.

Uses of Process Cost Systems

- Use to apply costs to ***similar*** products that are ***mass-produced*** in a ***continuous*** fashion.
- **Examples** include the production of cereal, paint, manufacturing steel, oil refining and soft drinks.

Illustration 21-1
Manufacturing processes

Use of Process Cost Systems

Examples of companies that primarily use either a process cost system or a job order cost system.

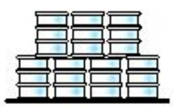
Process Cost System		Job Order Cost System	
Company	Product	Company	Product
Jones Soda, PepsiCo	Soft drinks 	Young & Rubicam, J. Walter Thompson	Advertising 
ExxonMobil, Royal Dutch Shell	Oil 	Disney, Warner Brothers	Movies 
Intel, Advanced Micro Devices	Computer chips 	Center Ice Consultants, Ice Pro	Ice rinks 
Dow Chemical, DuPont	Chemicals 	Kaiser, Mayo Clinic	Patient health care 

Illustration 21-2
Process cost and job order cost companies and products

Use of Process Cost Systems

Question

Which of the following items is *not* a characteristic of a process cost system:

a. Once production begins, it continues until the finished product emerges.

b. The focus is on continually producing homogenous products.

c. When the finished product emerges, all units have precisely the same amount of materials, labor, and overhead.

d. The products produced are heterogeneous in nature.

Process Cost for Service Companies

Service companies that provide individualized, non-routine services will probably benefit from using a **job order** cost system.

Those that perform routine, repetitive services will probably be better off with a **process cost** system.

Similarities and Differences Between Job Order Cost and Process Cost Systems

Job Order Cost

- Costs assigned to **each job**.
- Products have **unique characteristics**.

Process Cost

- Costs tracked through a **series of connected manufacturing processes or departments**.
- Products are **uniform** or relatively homogeneous and produced in a **large volume**.

Job Order Cost and Process Cost Flow

SIMILARITIES

1. Manufacturing cost elements.
2. Accumulation of the costs of materials, labor, and overhead.
3. Flow of costs.

DIFFERENCES

1. Number of work in process accounts used.
2. Documents used to track costs.
3. Point at which costs are totaled.
4. Unit cost computations.

Job Order Cost and Process Cost Flow

Feature	Job Order Cost System	Process Cost System
Work in process accounts	One work in process account	Multiple work in process accounts
Documents used	Job cost sheets	Production cost reports
Determination of total manufacturing costs	Each job	Each period
Unit-cost computations	Cost of each job ÷ Units produced for the job	Total manufacturing costs ÷ Equivalent units produced during the period

Illustration 21-4
Job order versus process cost systems

Use of Process Cost Systems

Question

Indicate which of the following statements is not correct:

a. Both a job order and a process cost system track the same three manufacturing cost elements – direct materials, direct labor, and manufacturing overhead.

b. In a job order cost system, only one work in process account is used, whereas in a process cost system, multiple work in process accounts are used.

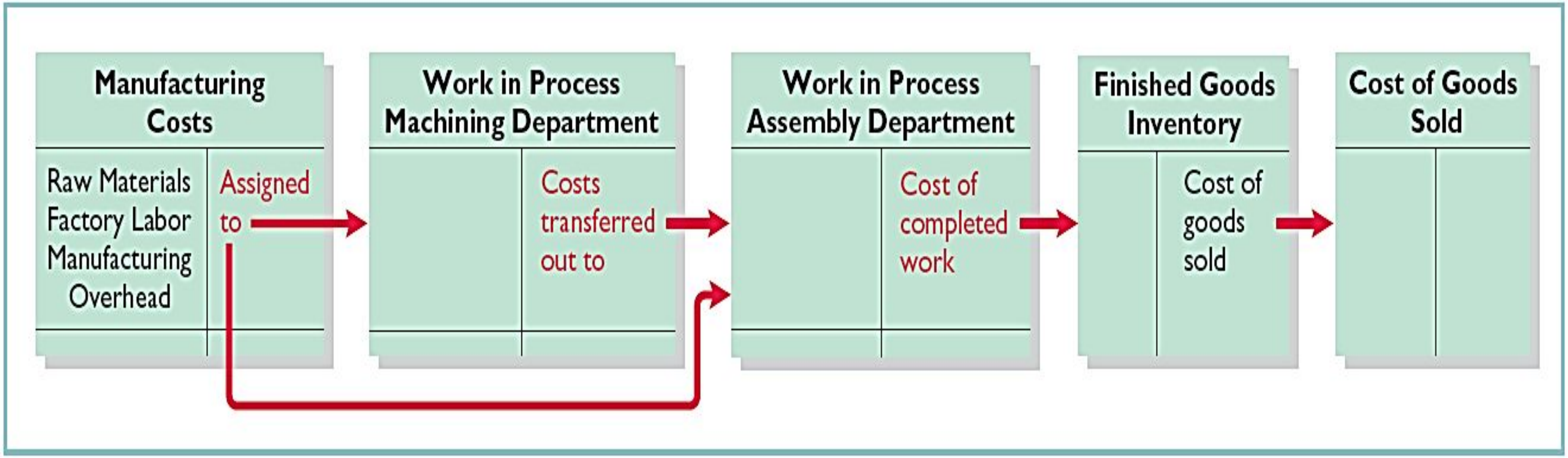
c. Manufacturing costs are accumulated the same way in a job order and in a process cost system.

d. Manufacturing costs are assigned the same way in a job order and in a process cost system.

2

Explain the flow of costs in a process cost system and the journal entries to assign manufacturing costs.

Process Cost Flow



Assigning Manufacturing Costs

- ❖ Debit Raw Materials Inventory for purchases of raw materials.
- ❖ Debit Factory Labor for factory labor incurred.
- ❖ Debit Manufacturing Overhead for the overhead cost incurred.

Assigning Manufacturing Costs (Journal entry to record materials used)

MATERIAL COSTS

Work in Process—Machining	XXXXX	
Work in Process—Assembly	XXXXX	
Raw Materials Inventory		XXXXX

FACTORY LABOR COSTS

Work in Process—Machining	XXXXX	
Work in Process—Assembly	XXXXX	
Factory Labor	XXXX	

MANUFACTURING OVERHEAD COSTS

Work in Process—Machining	XXXXX	
Work in Process—Assembly	XXXXX	
Manufacturing Overhead		XXXXX

Assigning Manufacturing Costs (TRANSFERS)

Monthly Entry to transfer goods to <u>next department</u>:

Work in Process—Assembly	XXXXX	
Work in Process—Machining		XXXXX

Entry to transfer completed goods to <u>Finished Goods</u>:

Finished Goods Inventory	XXXXX	
Work in Process—Assembly		XXXXX

Entry to record <u>Cost of Goods</u> sold at the time of sale:

Cost of Goods Sold	XXXXX	
Finished Goods Inventory		XXXXX

LEARNING OBJECTIVE **3**

Compute equivalent units.

Equivalent units of production measure the **work done during the period**, expressed in fully completed units. Companies use this measure to determine the **cost per unit of completed product**.

Illustration: Suppose you have a work-study job in the office of your college's president, and she asks you to compute the cost of instruction per full-time equivalent student at your college. The college's vice president for finance provides the following information.

Total cost of instruction	$9,000,000

Student population:

Full-time students	900
Part-time students	1,000

Compute Equivalent Units

Part-time students take 60% of the classes of a full-time student during the year. To compute the number of full-time equivalent students per year, you would make the following computation.

Full-Time Students	+	Equivalent Units of Part-Time Students	=	Full-Time Equivalent Students
	+		=	

Cost of instruction per full-time equivalent student =

Total cost of instruction $9,000,000

Number of full-time equivalent students ÷ 1,500

$ 6,000

Weighted-Average Method

- Considers the degree of completion (weighting) of units completed and transferred out and units in ending work in process. Most widely used method.
- Beginning work in process not part of computation of equivalent units.

Units Completed and Transferred Out	+	Equivalent Units of Ending Work in Process	=	Equivalent Units of Production

Illustration: The output of Kori Company's Packaging Department during the period consists of 10,000 units completed and transferred out, and 5,000 units in ending work in process, which are 70% completed.

Calculate the equivalent units of production.

Completed units 10,000

Work in process equivalent units (5,000 x 70%) 3,500

13,500

Refinements on the Weighted-Average Method

- Conversion costs are labor costs plus overhead costs.
- Beginning work in process is not part of the equivalent-units-of-production formula.

Units Completed and Transferred Out–Materials	+	Equivalent Units of Ending Work in Process–Materials	=	Equivalent Units of Production–Materials
Units Completed and Transferred Out–Conversion Costs	+	Equivalent Units of Ending Work in Process–Conversion Costs	=	Equivalent Units of Production–Conversion Costs

Refinements on Weighted-Average

Kellogg Company has produced Eggo® Waffles since 1970. Three departments produce these waffles: Mixing, Baking, and Freezing/Packaging. The Mixing Department combines dry ingredients, including flour, salt, and baking powder, with liquid ingredients, including eggs and vegetable oil, to make waffle batter. Information related to the Mixing Department at the end of June.

MIXING DEPARTMENT

		Percentage Complete	
	Physical Units	**Materials**	**Conversion Costs**
Work in process, June 1	100,000	100%	70%
Started into production	800,000		
Total units	900,000		
Units transferred out	700,000		
Work in process, June 30	200,000	100%	60%
Total units	900,000		

Solution

MIXING DEPARTMENT		
	Equivalent Units	
	Materials	Conversion Costs
Units transferred out	700,000	700,000
Work in process, June 30		
200,000 × 100%	200,000	
200,000 × 60%		120,000
Total equivalent units	900,000	820,000

DO IT! 3 Equivalent Units

The fabricating department, Outdoor Essentials, has the following production and cost data for the current month.

Beginning Work in Process	Units Transferred Out	Ending Work in Process
–0–	15,000	10,000

Materials are entered at the beginning of the process. The ending work in process units is 30% complete as to conversion costs. Compute the equivalent units of production for (a) materials and (b) conversion costs.

a) Units transferred out 15,000

Ending work in process units 10,000

25,000

b) Units transferred out 15,000

Equivalent unit in ending WIP (10,000 x 30%) 3,000

18,000

4 Complete the four steps to prepare a production cost report.

A production cost report is the

- Key document used to understand activities.
- Prepared for **each** department and shows Production Quantity and Cost data.
- **Four steps** in preparation:

Step 1: Compute physical unit flow

Step 2: Compute equivalent units of production

Step 3: Compute unit production costs

Step 4: Prepare a cost reconciliation schedule

Production Cost Report

Flow of costs to make an Eggo® Waffle and the related production cost reports for each department.

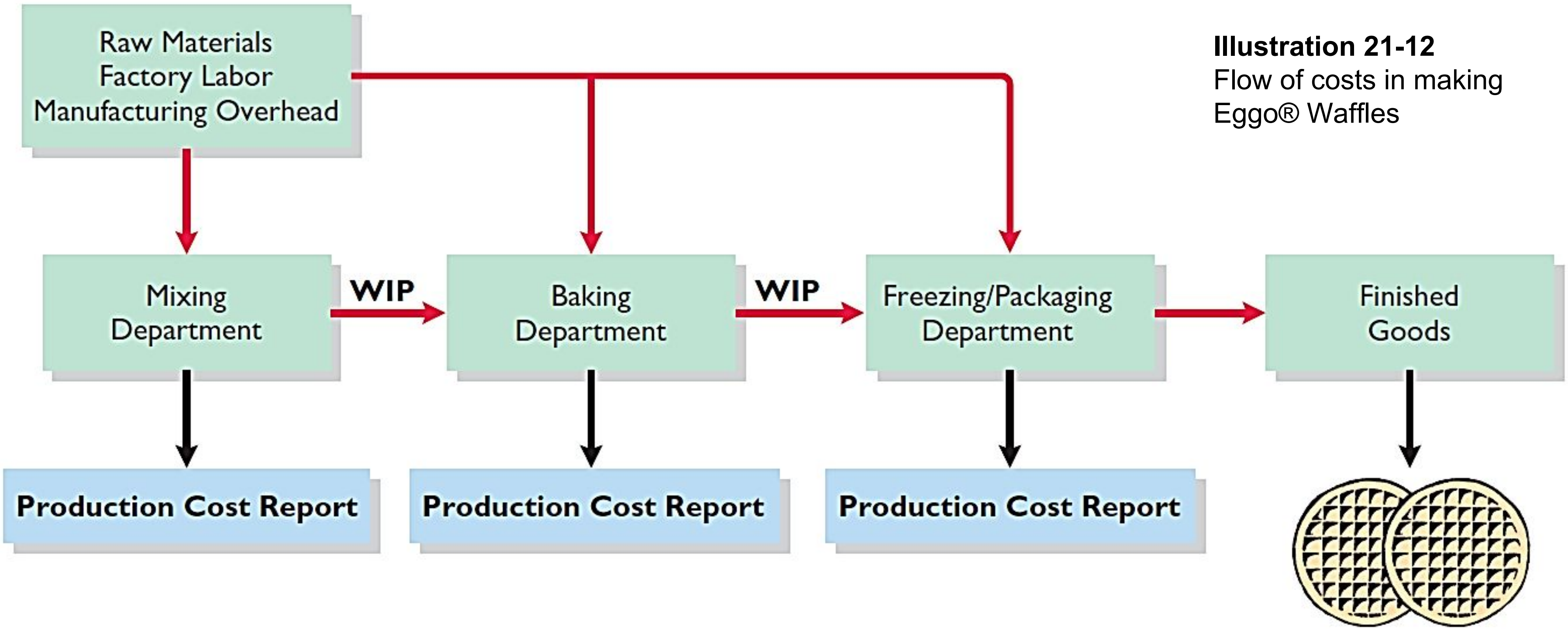


Illustration 21-12
Flow of costs in making Eggo® Waffles

MIXING DEPARTMENT

Units	
Work in process, June 1	100,000
Direct materials: 100% complete	
Conversion costs: 70% complete	
Units started into production during June	800,000
Units completed and transferred out to Baking Department	700,000
Work in process, June 30	200,000
Direct materials: 100% complete	
Conversion costs: 60% complete	
Costs	
Work in process, June 1	
Direct materials: 100% complete	$ 50,000
Conversion costs: 70% complete	35,000
Cost of work in process, June 1	$ 85,000
Costs incurred during production in June	
Direct materials	$400,000
Conversion costs	170,000
Costs incurred in June	$570,000

Illustration 21-13

Unit and cost data—Mixing Department

Production Cost Report

Compute the Physical Unit Flow (Step 1)

- **Physical units - actual units** to be accounted for during a period, regardless of work performed.
- **Total units to be accounted for - units started** (or transferred) into production during the period + **units in production at beginning** of period.
- **Total units accounted for - units transferred out** during period + **units in process at end** of period.

Production Cost Report

Compute the Physical Unit Flow (Step 1)

MIXING DEPARTMENT	
	Physical Units
Units to be accounted for	
Work in process, June 1	100,000
Started (transferred) into production	800,000
Total units	**900,000**
Units accounted for	
Completed and transferred out	700,000
Work in process, June 30	200,000
Total units	**900,000**

Illustration 21-14
Physical unit flow—Mixing Department

Production Cost Report

Compute Equivalent Units of Production (Step 2)

Mixing Department

- Department adds materials at beginning of process and
- Incurs conversion costs uniformly during the process.

	Equivalent Units	
	Materials	Conversion Costs
Units transferred out	700,000	700,000
Work in process, June 30		
200,000 × 100%	200,000	
200,000 × 60%		120,000
Total equivalent units	**900,000**	**820,000**

Illustration 21-15
Computation of equivalent units—Mixing Department

Production Cost Report

Compute Unit Production Costs (Step 3)

- Costs expressed in terms of equivalent units of production.
- When equivalent units of production are different for materials and for conversion costs, three unit costs are computed:
  1. Materials
  2. Conversion
  3. Total Manufacturing

Production Cost Report

Compute Unit Production Costs (Step 3)

Compute total materials cost related to Eggo® Waffles:

Illustration 21-16

Work in process, June 1	
Direct materials costs	$ 50,000
Cost added to production during June	
Direct material cost	400,000
Total material costs	**$450,000**

Illustration 21-17

Total Materials Cost	÷	Equivalent Units of Materials	=	Unit Materials Cost
	÷		=	

Production Cost Report

Compute Unit Production Costs (Step 3)

Compute total materials cost related to Eggo® Waffles:

Illustration 21-18

Work in process, June 1		
Conversion costs	$ 35,000	
Costs added to production during June		
Conversion costs	170,000	
Total conversion costs	$205,000	

Illustration 21-19

Total Conversion Costs	÷	Equivalent Units of Conversion Costs	=	Unit Conversion Cost
	÷		=	

LO 4

Production Cost Report

Compute Unit Production Costs (Step 3)

Illustration 21-19

Total Conversion Costs	÷	Equivalent Units of Conversion Costs	=	Unit Conversion Cost
$205,000	÷	820,000	=	$0.25

Compute total manufacturing costs per unit:

Illustration 21-20

Unit Materials Cost	+	Unit Conversion Cost	=	Total Manufacturing Cost per Unit
	+		=	

Production Cost Report

Prepare a Cost Reconciliation Schedule (Step 4)

Kellogg charged total costs of $655,000 to the Mixing Department in June, calculated as follows.

Illustration 21-21

Costs to be accounted for		
Work in process, June 1	$ 85,000	
Started into production	570,000	
Total costs		**$655,000**

Production Cost Report

Prepare a Cost Reconciliation Schedule (Step 4)

MIXING DEPARTMENT
Cost Reconciliation Schedule

Costs accounted for		
Transferred out (700,000 × \$0.75)		\$ 525,000
Work in process, June 30		
Materials (200,000 × \$0.50)	\$100,000	
Conversion costs (120,000 × \$0.25)	30,000	130,000
Total costs		**\$655,000**

Illustration 21-22
Cost reconciliation schedule—
Mixing Department

Prepare the Production Cost Report

MIXING DEPARTMENT
Production Cost Report
For the Month Ended June 30, 2017

		Equivalent Units		
	Physical Units	Materials	Conversion Costs	
Quantities				
Units to be accounted for	Step 1	Step 2		
Work in process, June 1	100,000			
Started into production	800,000			
Total units	900,000			
Units accounted for				
Transferred out	700,000	700,000	700,000	
Work in process, June 30	200,000	200,000	120,000	(200,000 × 60%)
Total units	900,000	900,000	820,000	
Costs				
Unit costs Step 3		Materials	Conversion Costs	Total
Total cost	(a)	$450,000	$205,000	$655,000
Equivalent units	(b)	900,000	820,000	
Unit costs [(a) ÷ (b)]		$0.50	$0.25	$0.75
Costs to be accounted for				
Work in process, June 1				$ 85,000
Started into production				570,000
Total costs				$655,000
Cost Reconciliation Schedule Step 4				
Costs accounted for				
Transferred out (700,000 × $0.75)				$525,000
Work in process, June 30				
Materials (200,000 × $0.50)			$100,000	
Conversion costs (120,000 × $0.25)			30,000	130,000
Total costs				$655,000

Illustration 21-23
Production cost report

Production Cost Report

Question

Largo Company has unit costs of $10 for materials and $30 for conversion costs. If there are 2,500 units in ending work in process, 40% complete as to conversion costs and fully complete as to materials cost, the total cost assignable to the ending work in process inventory is:

a. $45,000.

b. $55,000.

c. $75,000.

d. $100,000.

DO IT! 4 Cost Reconciliation Schedule

In March, Rodayo Manufacturing had the following unit production costs: materials $6 and conversion costs $9. On March 1, it had zero work in process. During March, Rodayo transferred out 12,000 units. As of March 31, 800 units that were 25% complete as to conversion costs and 100% complete as to materials were in ending work in process. Assign the costs to the units transferred out and in process.

Costs transferred out (12,000 x $15)		$180,000
Work in process, March 31		
Materials (800 x $6)	$4,800	
Conversion costs (200* x $9)	1,800	6,600
Total costs		$186,600

*800 x 25%

LEARNING OBJECTIVE **5** **Explain just-in-time (JIT) processing and activity-based costing (ABC).**

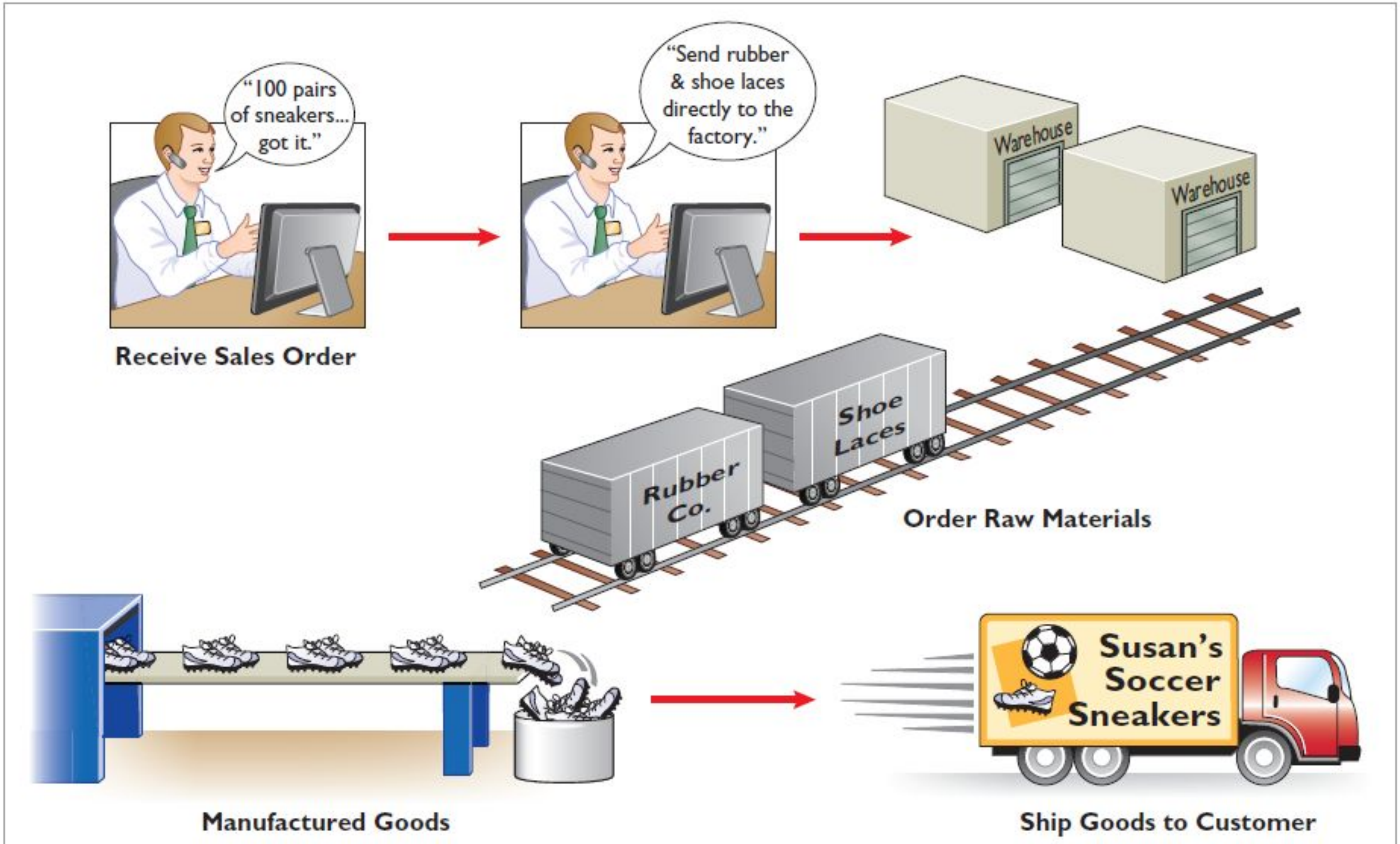
Just-In-Time Processing

A processing system that is dedicated to having the right products or parts as they are needed.

- **Objective:** To eliminate all manufacturing inventories to make funds and space available for more productive purposes.
- **Elements of JIT:** Dependable suppliers; multi-skilled workforce; total quality control system.
- **Benefits of JIT:** Significant reduction or elimination of manufacturing inventories; enhanced product quality; reduced rework and storage costs; savings from improved flow of goods.

Just-in-Time Processing

Illustration 21-24
Just-in-time processing



Activity-Based Costing

An overhead cost allocation system that focuses on activities performed in producing a product.

- **ABC System:** More that one basis of allocating activity costs to products is needed.
- **Assumptions of ABC:** All overhead costs related to the activity
  - must be driven by the cost driver used to assign costs to products.
  - should respond proportionally to changes in the activity level of the cost driver.

Activity-Based Costing

Activity	Cost Driver
Ordering raw materials	Ordering hours; number of orders
Receiving raw materials	Receiving hours; number of shipments
Materials handling	Number of requisitions; weight of materials; handling hours
Production scheduling	Number of orders
Machine setups	Setup hours; number of setups
Machining (fabricating, assembling, etc.)	Machine hours
Quality control inspections	Number of inspections
Factory supervision	Number of employees

Illustration 21-25
Activities and cost drivers in ABC

Management Insight

ABC Evaluated

Surveys of companies often show ABC usage of approximately 50%. Yet, in recent years, articles about ABC have expressed mixed opinions regarding its usefulness. To evaluate ABC practices and user satisfaction with ABC, a survey was conducted of 348 companies worldwide. Some of the interesting findings included the following: ABC methods are widely used across the entire value chain, rather than being primarily used to allocate production-specific costs; only 25% of non-ABC companies think they are accurately tracing the costs of activities, while 70% of ABC companies think their company does this well; and respondents felt that ABC provides greater support for financial, operational, and strategic decisions. More than 87% of respondents said that their ideal costing system would include some form of ABC. Since this significantly exceeds the 50% of the respondents actually using it, ABC usage may well increase in the future.

Source: William Stratton, Denis Desroches, Raef Lawson, and Toby Hatch, "Activity-Based Costing: Is It Still Relevant?" Management Accounting Quarterly (Spring, 2009), pp. 31–39.

DO IT! 5 JIT and ABC

Indicate whether the following statements are **true** or **false**.

1. Continuous process manufacturing often results in a reduction of inventory.

2. Companies that use just-in-time processing complete and store finished goods all the time to meet rush orders from customers.

3. A major benefit of just-in-time processing is production cost savings from the improved flow of goods through the processes.

DO IT! 5 JIT and ABC

Indicate whether the following statements are **true** or **false**.

4. An ABC system is similar to traditional costing systems in accounting for manufacturing costs but differs in regard to period costs.

5. The primary benefit of ABC is more accurate and meaningful costs.

6. In recent years, the amount of direct labor used in many industries has greatly increased and total overhead costs have significantly decreased.

LEARNING OBJECTIVE 6

APPENDIX 21A: Apply activity-based costing to a manufacturer.

Activity-Based Costing

Involves the following four steps.

1. Identify and classify the activities involved in the manufacture of specific products and assign overhead to cost pools.
2. Identify the cost driver that has a strong correlation to the costs accumulated in each cost pool.
3. Compute the activity-based overhead rate for each cost pool.
4. Allocate overhead costs to products, using the overhead rates determined for each cost pool.

Identify and Classify Activities and Allocate Overhead to Cost Pools (Step 1)

Overhead costs are assigned directly to the appropriate activity cost pool.

Illustration 21A-2
Activity cost pools and estimated overhead

ATLAS COMPANY	
Activity Cost Pools	Estimated Overhead
Manufacturing	$500,000
Setups	100,000
Purchase ordering	50,000
Product development	200,000
Property and plant	50,000
Total	$900,000

Identify Cost Drivers (Step 2)

Cost driver must accurately measure the actual consumption of the activity by the various products.

Illustration 21A-3: Cost drivers that Atlas Company identifies and their total expected use per activity cost pool.

Activity Cost Pools	Cost Drivers	Expected Use of Cost Drivers per Activity
Manufacturing	Machine hours	50,000 machine hours
Setups	Number of setups	2,000 setups
Purchase ordering	Number of purchase orders	2,500 purchase orders
Product development	Products developed	2 products developed
Property and plant	Square footage	25,000 square feet

Illustration 21A-3
Cost drivers and their expected use

Compute Activity-Based Overhead Rates (Step 3)

Next, the company computes an activity-based overhead rate per cost driver.

Illustration 21A-4

$$\frac{\textbf{Estimated Overhead per Activity}}{\textbf{Expected Use of Cost Drivers per Activity}} = \textbf{Activity-Based Overhead Rate}$$

Illustration 21A-5

	A	B	C	D	E	F
1	**ATLAS COMPANY**					
2	Activity Cost Pools	Estimated Overhead	÷	Expected Use of Cost Drivers per Activity	=	Activity-Based Overhead Rates
3	Manufacturing	$500,000		50,000 machine hours		$10 per machine hour
4	Setups	100,000		2,000 setups		$50 per setup
5	Purchase ordering	50,000		2,500 purchase orders		$20 per order
6	Product development	200,000		2 products developed		$100,000 per product
7	Plant and property	50,000		25,000 square feet		$2 per square foot
8	Total	$900,000				
9						

Allocate Overhead Costs to Products (Step 4)

In allocating overhead costs, it is necessary to know the expected use of cost drivers for each product. Because of its low volume and higher number of components, the Ab Coaster requires more setups and purchase orders than the Ab Bench.

			Expected Use of Cost Drivers per Product	
Activity Cost Pools	**Cost Drivers**	**Expected Use of Cost Drivers per Activity**	**Ab Bench**	**Ab Coaster**
Manufacturing	Machine hours	50,000 machine hours	30,000	20,000
Setups	Number of setups	2,000 setups	500	1,500
Purchase ordering	Number of purchase orders	2,500 purchase orders	750	1,750
Product development	Products developed	2 products developed	1	1
Property and plant	Square feet	25,000 square feet	10,000	15,000

Illustration 21A-6
Expected use of cost drivers per product

Allocate Overhead Costs to Products

To allocate overhead costs, Atlas multiplies the activity-based overhead rates per cost driver (Ill. 21A-5) by the number of cost drivers expected to be used per product (Ill. 21A-6).

	Ab Bench				
Activity Cost Pools	Expected Use of Cost Drivers per Product	×	Activity-Based Overhead Rates	=	Cost Allocated
Manufacturing	30,000		$10		$300,000
Setups	500		$50		25,000
Purchase ordering	750		$20		15,000
Product development	1		$100,000		100,000
Property and plant	10,000		$2.00		20,000
Total costs allocated (a)					$460,000
Units produced (b)					25,000
Overhead cost per unit [(a)÷(b)], rounded					$18.40

Illustration 21A-7
Allocation of activity cost pools to products

Allocate Overhead Costs to Products

To allocate overhead costs, Atlas multiplies the activity-based overhead rates per cost driver (Ill. 21A-5) by the number of cost drivers expected to be used per product (Ill. 21A-6).

		Ab Coaster				
Activity Cost Pools	Cost Allocated	Expected Use of Cost Drivers per Product	×	Activity-Based Overhead Rates	=	Cost Allocated
Manufacturing	$300,000	20,000		$10		$200,000
Setups	25,000	1,500		$50		75,000
Purchase ordering	15,000	1,750		$20		35,000
Product development	100,000	1		$100,000		100,000
Property and plant	20,000	15,000		$2.00		30,000
Total costs allocated (a)	$460,000					$440,000
Units produced (b)	25,000					5,000
Overhead cost per unit [(a)÷(b)], rounded	$18.40					$88.00

Illustration 21A-7

Comparing Unit Costs

	Ab Bench		Ab Coaster	
Manufacturing Costs	**Traditional Costing**	**ABC**	**Traditional Costing**	**ABC**
Direct materials	$40.00	$40.00	$30.00	$ 30.00
Direct labor	12.00	12.00	12.00	12.00
Overhead	30.00	18.40	30.00	88.00
Total direct cost per unit	**$82.00**	**$70.40**	**$72.00**	**$130.00**
	Overstated $11.60		**Understated $58.00**	

Illustration 21A-8
Comparison of unit product costs

Likely consequence of differences in assigning overhead.

- Overpricing the Ab Bench and possibly losing market share to competitors.
- Sacrificing profitability by underpricing the Ab Coaster.

Benefits and Limitations of ABC

Benefits

1. More cost pools, therefore more accurate product costing.
2. Enhanced control over overhead costs.
3. Better management decisions.

Limitations

1. Expensive to use.
2. Some arbitrary allocations remain.